

## Session 3

# Networking Fundamentals

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*Understanding how devices talk to each other across a network*



Lecture by

**Shamali Kene**

Project Engineer

# Topics Covered

01

## TCP/IP Protocol Framework

The rules that govern how data travels across a network

02

## IP Address

The unique digital address assigned to every device

03

## Subnet Masks

Dividing an IP address into network and host portions

04

## CIDR Notation

Flexible, efficient allocation of IP address blocks

# TCP/IP Protocol Framework

## What is TCP/IP?

**TCP/IP** is the framework that defines how data moves from one device to another across networks.

*Transmission Control Protocol / Internet Protocol*

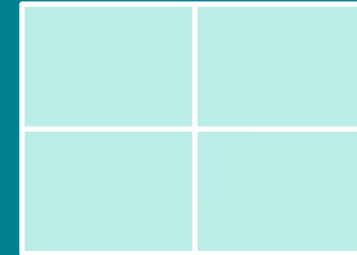
It is the set of rules computers use to communicate over a network or the Internet.

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**Think of it like sending a parcel:**

**IP** → Where should the parcel go?

**TCP** → Make sure the parcel arrives correctly and in order.



PARCEL

*IP addresses it. TCP delivers it —  
complete, correct, in order.*

TCP/IP =

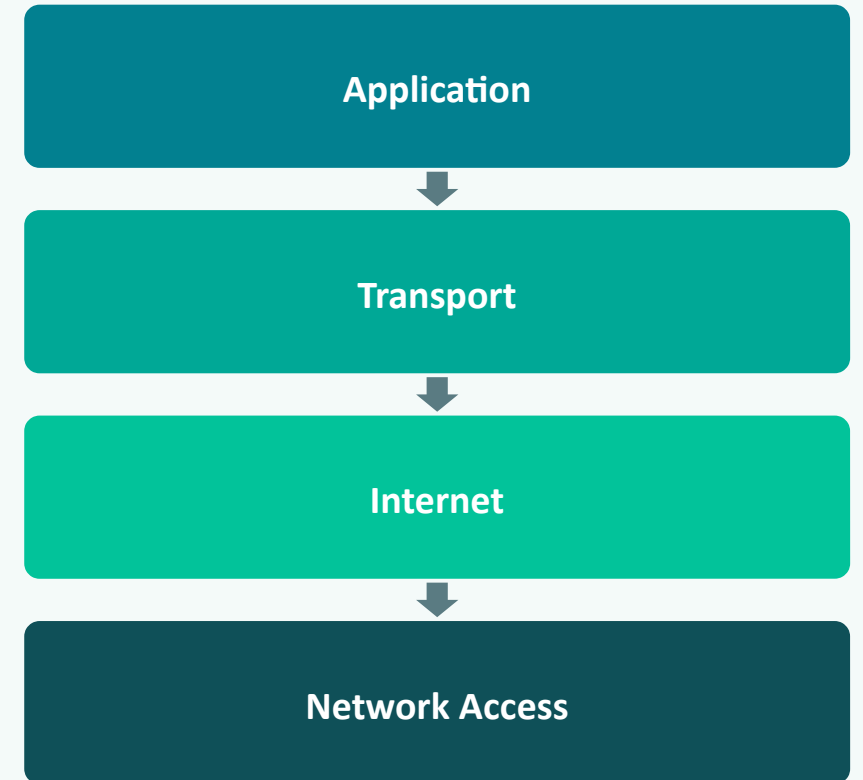
**Transmission Control Protocol /  
Internet Protocol**

# TCP/IP Has 4 Layers

| Layer          | What it does                | Examples            |
|----------------|-----------------------------|---------------------|
| Application    | Services applications use   | HTTP, DNS, FTP, SSH |
| Transport      | End-to-end communication    | TCP, UDP            |
| Internet       | Addressing & routing        | IP, ICMP            |
| Network Access | Sends data over the network | Ethernet, Wi-Fi     |

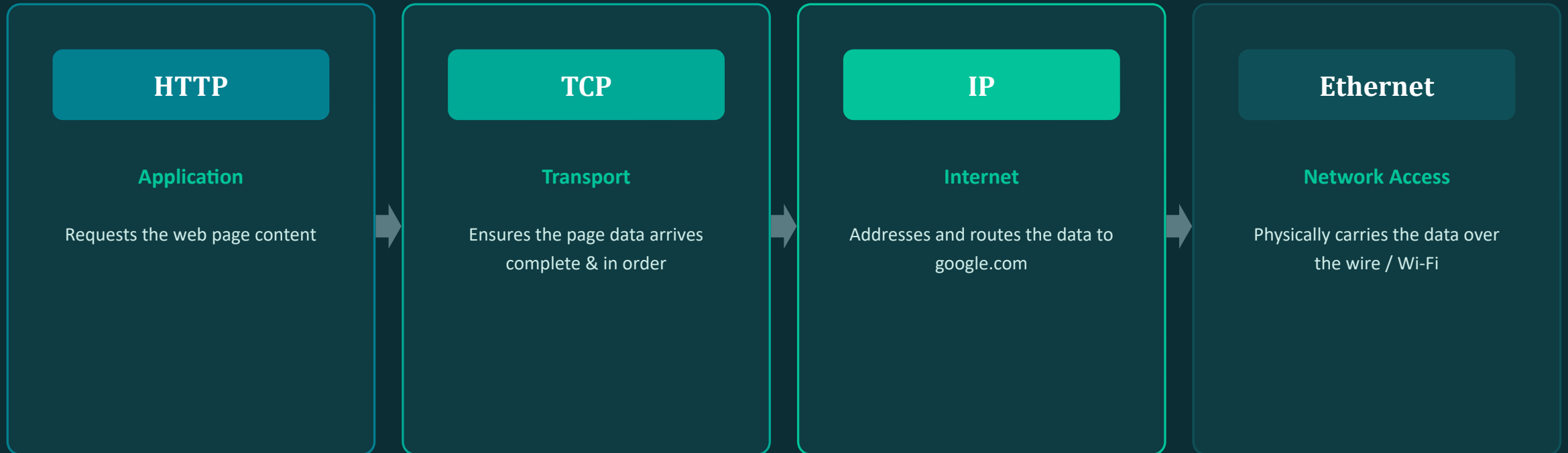
*Data flows down to send, and back up to receive.*

Easy way to remember



# A Real Example

*Opening a website walks through every layer of TCP/IP*



# IP Addresses

## What is an IP Address?

An IP address (Internet Protocol address) is a unique label assigned to every device connected to a computer network.

It acts like a digital home address, allowing devices to find and talk to each other.



**192.168.1.10**

*Its unique digital address*

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Every device on a network — phones, laptops, servers  
— has one.

# Types of IP Addresses

## Public IP

Assigned by your internet provider to your entire home or office network. Visible to the whole internet and allows you to access websites.

## Private IP

Assigned by your router to individual devices (phone, laptop) inside your local network. Keeps devices safe by hiding them from the public internet.

## Dynamic IP

Temporary addresses that change automatically over time. Your internet provider rotates them to save resources — most household devices use these.

## Static IP

Permanent addresses that never change. Manually set for servers, business networks, or websites needing a constant, reliable connection.

# IPv4 vs IPv6

## IPv4

**32**

bit address

**4.3 Billion**

unique addresses

Format example

**192.168.1.1**

Disadvantage

Limited number of addresses (only 4.3 billion); not secure.

## IPv6

**128**

bit address

**340 Undecillion**

unique addresses

Format example

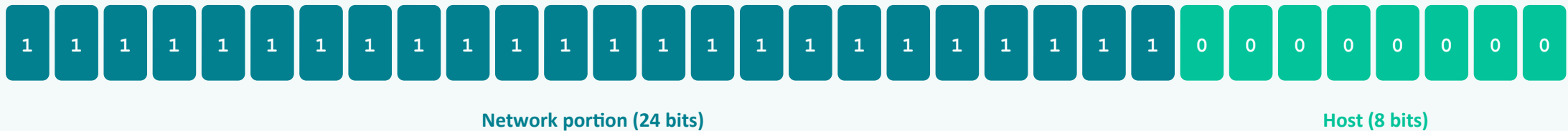
**2001:db8::1**

Disadvantage

Long, hard-to-remember address format.

# Subnet Masks

A **subnet mask** is a 32-bit number used with an IP address to divide it into two parts: the **network ID** and the **host ID**.



**Binary 1's** → represent the network portion of the IP address.

**Binary 0's** → represent the host portion.

*The subnet mask helps a device determine whether another device is on the same local network or a different network — deciding whether communication is direct or must go through a router.*

# Classless Inter-Domain Routing (CIDR)

**CIDR** is a method of IP address allocation and routing that allows more efficient use of IP addresses — allocating based on a network **prefix** rather than a fixed class (A, B, or C).

## Why CIDR? — Classful addressing wastes IP addresses

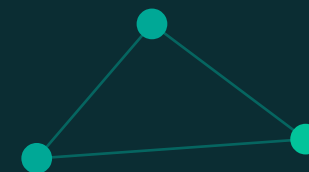
| Class | IPs Available | Hosts        | Example Wastage                    |
|-------|---------------|--------------|------------------------------------|
| A     | $2^{24}$      | $2^{24} - 2$ | Too large for small orgs           |
| B     | $2^{16}$      | $2^{16} - 2$ | Wastes 49,150 hosts for 214 needed |
| C     | $2^8$         | $2^8 - 2$    | Small networks only                |

### Problem

Organizations often need a host count that doesn't match class sizes — leading to wastage.

### Solution

CIDR allows flexible block allocation matching the exact number of addresses an organization requires — eliminating the fixed-class wastage.



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# Thank You

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Shamali Kene | Project Engineer

